



贝叶斯网络与不良结果途径 (AOP) 网络之间的关系以及如何使用它预测药物毒性

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将不良结果途径 (AOPs) 与贝叶斯网络 (BNs) 结合起来引起了广泛关注, 因为它们都使用有向无环图 (DAGs) 进行表示。然而, 目前尚未通过实验证实AOP网络在数学上是否与BNs一致。此外, 对于BNs的重要属性, 如马尔可夫毯 (Markov blankets), 尚未得到强调, 这错失了简化和优化模型的机会。在此, 我们总结了AOP网络与BNs之间的联系, 并探讨了其对毒性建模的影响。我们还展示了一个与药物相关的肝毒性案例研究。我们的结果证实, AOP网络在数学上与BNs一致, 并且引入BN的数学属性显著简化并提高了模型的效率。

关键词: 不良结局路径 途径; 贝叶斯网络; 药物开发; 体外检测; 肝毒性; 毒性预测

介绍
慢性动物试验仍被认为是毒性测试的黄金标准, 无论是在环境化学品评估还是药物开发中。然而, 现在人们普遍认识到, 基于动物的测试存在重要的局限性。在药物发现过程中, 动物试验经常无法预测药物候选物对人体的毒性。即使能够检测到毒性, 动物试验也常常只能提供对相关生物通路的有限见解。基于体外实验的测试可以直接以人体生物学作为毒性反应的依据, 因此其结果与人体生物学更相关, 且解释更明确。部分用体外实验替代动物测试, 也符合社会上减少毒性评估中动物使用的需求。已有大量文献以及资源可用于开发可用的分子实验。

开发毒性预测模型。^{(p2),(p5),(p6),(p7),(p8),(p9)}然而, 尽管在开发用于预测毒性的机器学习模型方面投入了大量精力, ^{(p10),(p11),(p12)} 如何开发适用于实际使用的准确且高效的模型仍然是一个重大挑战。

AOP 框架有潜力促进基于体外实验的毒性预测模型的开发。AOPs^{(p13)、(p14)} 和 AOP 网络是用于高效存储和访问评估毒性所需的机制知识的管理框架。AOPs 可以用于帮助化学品的毒性评估, 包括定性和定量评估。对于定量评估, 由于 AOPs 与 BN 共享使用 DAG 的表示方式, 结合 AOPs 与 BN 的研究引起了极大兴趣。由于 BNs 在概率推理方面有许多优势, ^(p15) 报告显示, 通过将 AOP 网络与 BN 结合, 在毒性推理方面取得了成功。

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The connection between Bayesian networks and adverse outcome pathway (AOP) networks and how to use it for predicting drug toxicity

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There is significant interest in combining adverse outcome pathways (AOPs) with Bayesian networks (BNs) because of their shared representation using directed acyclic graphs (DAGs). However, it has not been verified empirically whether AOP networks are mathematically congruent with BNs. Furthermore, important properties for BNs, such as Markov blankets, have not been emphasized, which is a missed opportunity for simplifying and optimizing the model. Here, we summarize the connection between AOP networks and BNs and explore the implications for toxicity modeling. We also present a case study in drug-related liver toxicity. Our results confirm that AOP networks are congruent mathematically with BNs, with incorporation of the mathematical properties of BN leading to significantly simplified and more efficient models.

Keywords: adverse outcome pathway; Bayesian networks; drug development; in vitro assays; hepatotoxicity; toxicity prediction

Introduction
Chronic animal testing is still considered the gold standard of toxicity testing, in both the evaluation of environmental chemicals and drug development. However, it is now widely recognized that animal-based testing has important shortcomings. In the drug discovery process, animal testing often fails to predict the human toxicity of drug candidates. Even when toxicity can be detected, animal testing often can only provide limited insight into the biological pathways that are involved. Testing based on *in vitro* assays^{(p1),(p2)} can be directly anchored by human biology for toxicity responses; thus, the results are more relevant to human biology with clear interpretations. Partially replacing animal-based testing with *in vitro* assays is also consistent with societal demand for reducing animal use in toxicity evaluations.^{(p3),(p4)} There is a significant literature on, as well as resources for, developing molecular assays that could be used

to develop predictive models for toxicity.^{(p2),(p5),(p6),(p7),(p8),(p9)} Nevertheless, although there is substantial focus on developing machine learning models to predict toxicity,^{(p10),(p11),(p12)} how to develop accurate and efficient models suitable for practical use remains a major challenge.
The AOP framework has the potential to facilitate the development of predictive models for toxicity with *in vitro* assays. AOPs^{(p13),(p14)} and AOP networks are knowledge management frameworks for efficiently storing and accessing mechanistic knowledge required for assessing toxicity. AOPs can be utilized to aid toxicity evaluations for chemicals both qualitatively and quantitatively. For quantitative assessment, there is significant interest in combining AOPs with BNs because of their shared representation using DAGs. With many advantages of BNs in probability reasoning,^(p15) success has been reported in performing inference for toxicity by combining AOP networks with BN

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建模.^{(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23)} 然而，作为统计模型，BN的应用需要重要的数学假设。尽管AOP网络和BN共享DAG的拓扑结构，但AOP网络在数学上是否与BN一致尚未通过实验证实。此外，即使在使用BN进行基于AOP的建模的研究中，一些重要特性，如马尔可夫毯(Markov blanket)，在模型开发过程中也未被强调，这错失了简化和优化模型的机会。马尔可夫毯是指最小节点集合，如果知道这些节点的状态，就能使目标节点在条件上与网络中的所有其他节点独立。它可以被视为一个将节点内部状态与外部环境分开的茧。正如我们下面讨论的，这一特性在进行推理时非常有用。

在这里，我们首先总结了 AOP 网络与 BN 之间的联系，并探讨其对毒性建模的影响。我们介绍了 BN 中的重要概念（马尔可夫包和 d-分离），这些概念有助于开发优化的模型。我们通过一个药物肝毒性潜能预测的案例研究来说明这些概念。我们的结果证实，AOP 网络在数学上与 BN 一致，并且结合 BN 的数学特性能够显著简化模型，提高肝毒性潜能预测的效率。

AOP 和 AOP 网络

AOP 是一种分析性构想，用于描述在不同生物组织水平上因果关联事件的连续链条，这些事件会导致个体或群体层面的不良影响^{(p13),(p14)} (图 1)。^(p23) AOP 是正在构建的毒理学知识框架的核心要素，该框架旨在通过机制推理来支持化学风险评估。AOP 的描述和构建需要考虑生物学合理性以及现有证据，说明生物系统的扰动如何通过各种步骤导致特定的不良结果（更多详情请参见 <https://aopkb.org/> 和 <https://aopwiki.org/>）。在其基本形式中，AOP 以事件的线性链条呈现，从分子起始事件（MIE）开始，通过一系列关键事件（KEs），最终导致顶端的不良结果。对于图 1 中的示例，芳基羧酸受体（AHR）激活是 MIE，它导致 CYP1A2/5 的诱导，进而导致尿卟啉的变化。

最终，这一系列关键事件（KEs）以尿卟啉症告终，这是在有机体水平上的不良结果。AOP的线性构建非常适合传递机制性知识，并有助于确定作用方式及开发体外测试。

鉴于生物过程通常非常复杂，多个相关的AOP不可避免地会共享一些关键事件。这自然导致了各种复杂程度的AOP网络，如胆汁淤积的AOP网络所示，其中的子集在图2a中有所详细说明。AOP网络的一个重要特性是它们以有向无环图（DAG）表示。在图论中，DAG由顶点（节点）和边（弧）组成，每条边从一个顶点指向另一个顶点，并且沿着这些方向永远不会形成闭环。在AOP网络中，每个关键事件（KE）可能由一个或多个前置KE（或初始分子事件，MIE）引起，并可以导致一个或多个后续KE，如边或弧（关键事件关系，KER）的方向所示。如果从一个MIE出发并沿着KER的方向前进，最终会在个体或群体水平上到达不良结果。用DAG表示的这种机制知识与通常的人类推理方式是一致的。

AOP框架已经从一个主要的概念性构想发展为一个越来越实用的工具，在风险评估和监管决策中有许多应用。然而，对AOP框架应用的全面调查超出了本综述的范围。不过，在结束本节之前，我们除案例研究中讨论的肝毒性示例外，还强调了AOP网络应用的一些有趣示例。

鉴于内分泌系统调节组织和器官的发育与功能，内分泌干扰物（EDs）可能导致严重的健康影响。^{(p28),(p29)} 然而，由于内分泌系统涉及多种激素和信号通路，构建EDs的AOPs（不良结果途径）受到极大关注。Zilliacus等人^(p30) 组建了一个包含42条AOPs的网络，这些AOPs与因雌激素、雄激素及类固醇生成途径扰动而导致的哺乳动物生殖毒性相关。他们进一步识别了16个子网络，这些子网络涉及激素水平或激素活性、癌症结果、男性和女性生殖系统以及总体生育能力的影响。

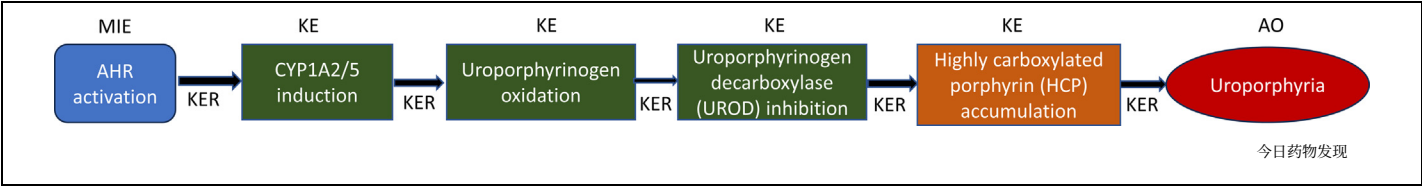


图 1
不良结果途径（AOP）：芳基羧酸受体（AHR）激活导致尿卟啉症。该AOP在AOPwiki中列为AOP 131。缩写：AO，不良结果；KE，关键事件；KER，关键事件关系；MIE，分子起始事件。

modeling.^{(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23)} However, the application of BNs as statistical models requires important mathematical assumptions. Although AOP networks and BNs share the topological structure of DAGs, whether AOP networks are mathematically congruent with BN has not been verified empirically. Furthermore, even with studies using BNs for AOP-based modeling, important properties, such as Markov blankets, have not been emphasized in the model development process, which is a missed opportunity for simplifying and optimizing the model. A Markov blanket is the minimal set of nodes that, if their states are known, renders a target node conditionally independent of all other nodes in the network. It can be considered as a cocoon separating the state of the node inside from its external environment. This property is extremely useful for making inference, as we discuss below.

Here, we first summarize the connection between AOP networks and BNs and explore the implications for toxicity modeling. We introduce the important concepts in BN (Markov blanket and d-separation), which will be beneficial for developing optimized models. We illustrate these concepts with a case study in the prediction of drug hepatotoxicity potential. Our results confirm that AOP networks are mathematically congruent with BNs, and that the incorporation of the mathematical properties of BNs leads to significantly simplified and more efficient models for the prediction of hepatotoxicity potential.

AOP and AOP networks

AOP is an analytical construct that describes a sequential chain of causally linked events at different levels of biological organization, which lead to an adverse effect at individual or population level^{(p13),(p14)} (Figure 1).^(p23) AOPs are the central element of a toxicological knowledge framework being built to support chemical risk assessment based on mechanistic reasoning. AOPs are described and constructed by considering biological plausibility and available evidence for how a perturbation of the biological system can lead, through various steps, to a specific adverse outcome (see <https://aopkb.org/> and <https://aopwiki.org/> for more details). In its basic form, an AOP is presented as a linear chain of events, from the molecular initiating event (MIE), through a sequence of key events (KEs), leading to the apical adverse outcome. For the example in Figure 1, aryl hydrocarbon receptor (AHR) activation is the MIE, which leads to CYP1A2/5 induction, which, in turn, leads to uroporphyrinogen oxidation, and so on.

Eventually, this chain of KEs concludes in uroporphyria, the adverse outcome at the organism level. The linear construction of AOPs is perfectly suited to convey mechanistic knowledge and assist in determining the mode of action and developing *in vitro* tests.

Given that biological processes are usually very complex, multiple related AOPs inevitably share some key events. This naturally leads to AOP networks of various complexity, as illustrated by the AOP network for cholestasis, subsets of which are detailed in Figure 2a).^(p24) One important property of AOP networks is that they are represented as DAGs. In graph theory, a DAG comprises vertices (nodes) and edges (arcs), with each edge leading from one vertex to another, such that following those directions will never form a closed loop. In AOP networks, each KE might be caused by one or more preceding KEs (or MIEs) and can lead to one or more subsequent KEs, as indicated by the direction of edges or arcs (key event relationships or KERs). If one starts from a MIE and follows the direction of KERs, one will eventually arrive at the adverse outcome at the individual or population level. This representation of mechanistic knowledge with DAGs is consistent with the usual human reasoning process for causal relationships. It also provides a natural connection with the formalism of BNs, which we touch on again below.

The AOP framework has matured from a largely conceptual construct to an increasingly practical tool with many applications in risk assessment and regulatory decision-making.^{(p25),(p26),(p27)} However, a comprehensive survey of the application of the AOP framework is beyond the scope of this review. Nevertheless, before concluding this section, we highlight interesting examples of the application of AOP networks in addition to the example for liver toxicity, which is discussed in the case study.

Given that the endocrine system regulates tissue and organ development and function, endocrine disruptors (EDs) can lead to serious health effects.^{(p28),(p29)} However, because the endocrine system involves many different hormones and signaling pathways, constructing AOPs for EDs has received significant interest. Zilliacus *et al.*^(p30) assembled an AOP network for 42 AOPs related to mammalian reproductive toxicity arising from disruption to the estrogen, androgen, and steroidogenesis modalities. They further identified 16 subnetworks related to effects on hormone levels or hormone activity, cancer outcomes, male and female reproductive systems, and overall effects on fertility and

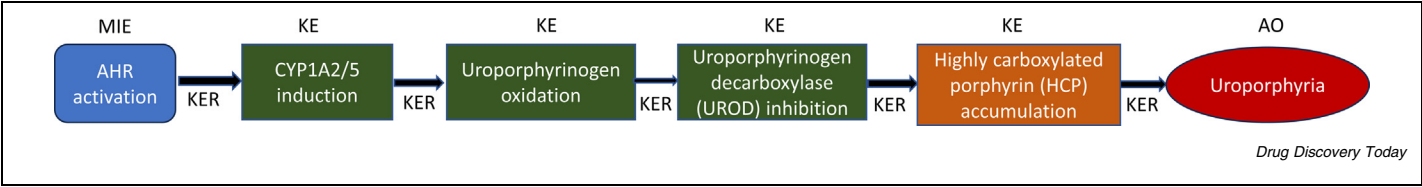


FIGURE 1
Adverse outcome pathway (AOP): aryl hydrocarbon receptor (AHR) activation leading to uroporphyria. This AOP is listed as AOP 131 in AOPwiki. Abbreviations: AO, adverse outcome; KE, key event; KER, key event relationship; MIE, molecular initiating event.

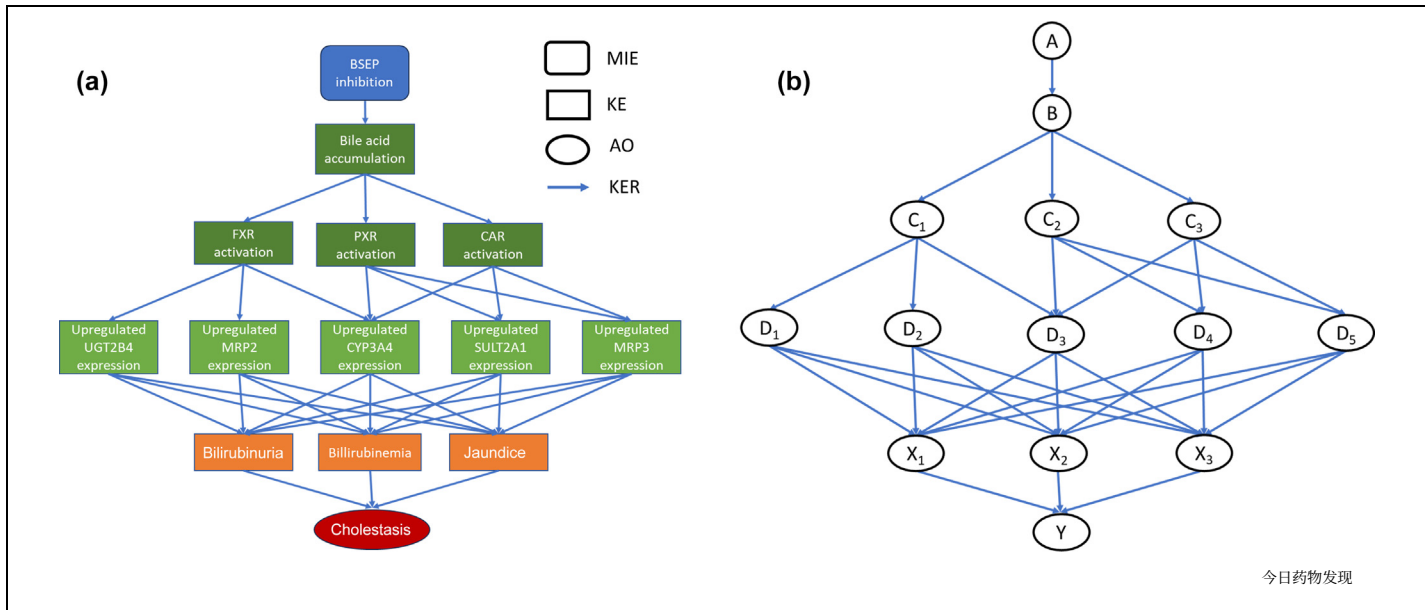


图2
用于不良结果通路（AOP）网络及相应贝叶斯网络（BN）的示例说明。（a）胆汁淤积的示例性AOP网络。该图仅包含胆汁淤积事件的子集，仅用于说明目的。它展示了有向无环图（DAG）的结构。（b）与(a)中的AOP网络具有相同结构的BN。节点为了在后文讨论中简洁起见进行了重新标注。缩略语：AO，不良结果；KE，关键事件；KER，关键事件关系；MIE，分子起始事件。

繁殖。总共有26个关键事件（KEs）和11个关键事件关系（KERs）被确定为AOP网络中的核心知识模块，这可以支持内分泌干扰物（EDs）的识别和评估。

Chai 等人利用比较毒理基因组数据库（CTD）和 AOP 维基的数据，构建了一个针对无机砷诱导的成年男性生殖功能损伤的 AOP 网络，该网络包含16条AOP。这提供了网络中关键事件（KEs）的全局视图，并突出了睾酮下降、反应性氧物种响应和精子发生的核心作用。它还帮助填补了该领域现有的知识空白，为未来的实验验证提供依据。

在神经毒理学方面，Spinu 等人描述了一个整合了九条线性 AOP 的 AOP 网络。作者确定了几个关键的关键事件（KE），其中细胞损伤/死亡被认为是网络中超链接最多的关键事件。所构建的网络有助于更好地理解发育期及成人/衰老期神经毒性相关的信号通路，并可为指导体外测试的发展提供基础，从而支持危害特征化和化学品风险评估。

最后，计算方法在辅助AOP网络的构建、分析和可视化方面取得了进展。^{(p33),(p34),(p35)}

贝叶斯网络

贝叶斯网络（BNs）是一类图模型，它能够将一组随机变量之间的概率依赖关系简化表示为有向无环图（DAG）。在这里，我们介绍基本概念，展示不是严格的数学化论述。关于贝叶斯网络的详细讨论可以在其他地方找到。^{(p36),(p37)} 图由一组节点或顶点以及节点对之间的连接组成，这些连接称为弧、边，或者非正式称为箭头。图2b

详细说明了这样的图，该图具有与图 2a 中的 AOP 网络相同的图形结构，但为了我们之后讨论的清晰而重新标注。在贝叶斯网络（BNs）中，图的每个节点对应一个随机变量。如果沿着弧的方向查看，图中不应存在环（有向无环图）。如果两个节点通过一条边相连，根据边的方向，它们形成一个父-子对。例如，在图 2b 中，节点 D₃ 的父节点是 C₁、C₂ 和 C₃，其子节点为 X₁、X₂ 和 X₃。同样，即使节点之间没有直接的边，位于同一有向路径上的节点也可以是祖先或后代。在图 2b 中，节点 A 和 B 是 D₃ 的祖先，尽管它们不是父节点。类似地，Y 是 D₃ 的后代，但不是子节点。

BN的一个基本属性是马尔可夫性。如果我们将BN的DAG中的所有节点标记为 $Z = \{Z_1, Z_2, \dots, Z_p\}$ ，则 P_z 的联合概率分布可以写成方程(1)：

$$P_z(Z) = \prod_{i=1}^p P_{Z_i}\{Z_i|PA(Z_i)\} \quad (1)$$

在这里，我们使用 $PA(Z_i)$ 来表示 Z_i 的父节点集合。这个属性大大便利了关于图中任意变量子集的概率计算。

鉴于 AOP 网络和 BN 都具有 DAG 结构，自然可以在 AOP 网络上构建 BN，并利用它们根据关于 AOP 中关键事件（KE）的测量来预测不良结果。AOP 提示了上游关键事件与下游关键事件或结果之间的因果关系。在 BN 中，联合概率分布表示为子节点在其父节点条件下的条件分布的乘积（公式（1））。如果 AOP 的因果关系是有效的，那么假设下游关键事件的概率分布也是合理的。

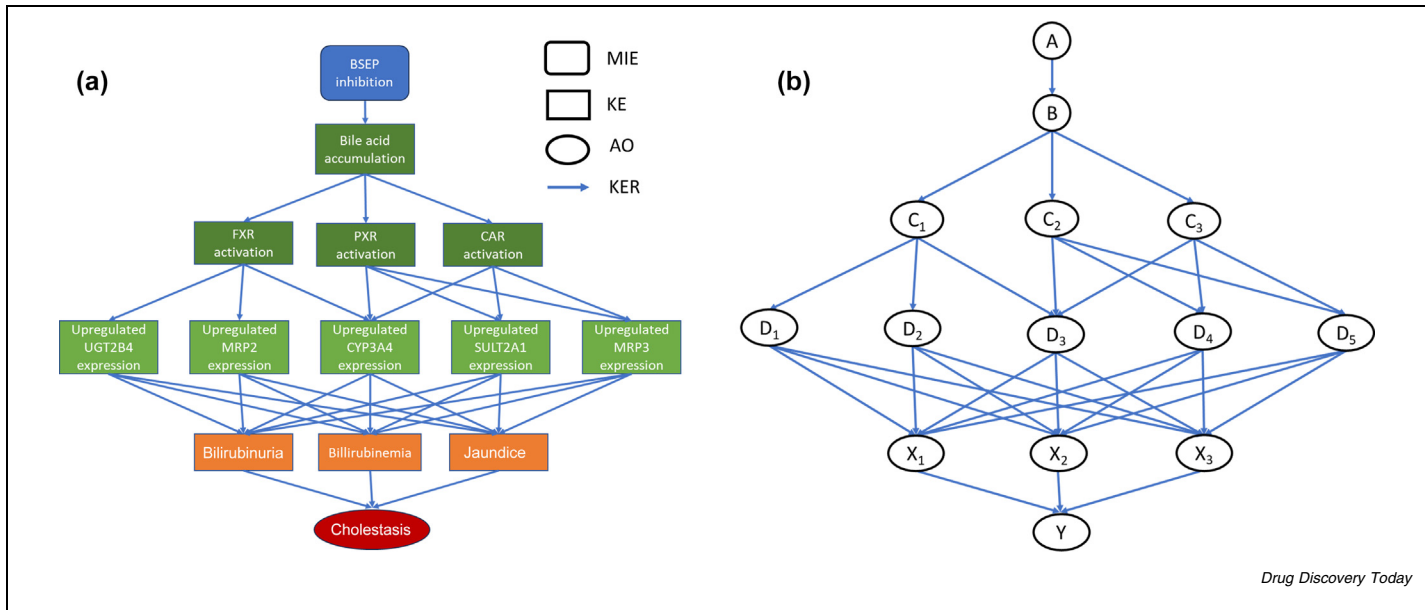


FIGURE 2
An illustrational example for adverse outcome pathway (AOP) networks and the corresponding Bayesian network (BN). **(a)** Illustrational AOP network for cholestasis. The graph only contains a subset of events for cholestasis and is intended for illustration purpose only. It demonstrates the structure of a directed acyclic graph (DAG). **(b)** A BN with the same structure as the AOP network in (a). Nodes are relabeled for conciseness in later discussion in the main text. Abbreviations: AO, adverse outcome; KE, key event; KER, key event relationship; MIE, molecular initiating event.

reproduction. In total, 26 KEs and 11 KERs were identified as core blocks of knowledge in the AOP network, which could support the identification and assessment of EDs.

Drawing on data from the comparative toxicogenomics database (CTD) and AOP wiki, Chai *et al.*^(p31) constructed a AOP network for inorganic arsenic-induced adult male reproductive impairment with 16 AOPs. This provided global views of KEs in the network and highlighted the essential roles of testosterone reduction, response to reactive oxygen species, and spermatogenesis. It also helped to address existing knowledge gaps in this area for future experimental verification.

In terms of neurotoxicology, Spinu *et al.*^(p32) described an AOP network integrating nine linear AOPs. The authors identified several critical Kes, with cell injury/death established as the most hyperlinked KE across the network. The derived network enabled better understanding of signaling pathways involved in developmental and adult/ageing neurotoxicity and could provide abasis to guide the development of *in vitro* tests supporting hazard characterization and chemical risk assessment.

Finally, there have been advances in computational methods to assist the construction, analysis, and visualization of AOP networks.^{(p33),(p34),(p35)}

Bayesian networks

BNs are a class of graphical models that enable simplified representation of probabilistic dependencies between a set of random variables as a DAG. Here, we introduce the basic concepts, and the presentation is not mathematically rigorous. Detailed discussions of BNs can be found elsewhere.^{(p36),(p37)} A graph comprises a set of nodes or vertices and a set of connections between pairs of vertices called arcs, edges, or, informally, arrows. Figure 2b

details such a graph, which has the same graphical structure as the AOP network in Figure 2a, but relabeled for the clarity of our subsequent discussion. In BNs, each node of the graph corresponds to a random variable. If one follows the directions of the arcs, there should be no loops (directed acyclic graph). If two nodes are connected by an edge, depending on the direction of the edge, they form a parent-child pair. For example, in Figure 2b, node D₃ has parents C₁, C₂, and C₃ and has children X₁, X₂, and X₃. Similarly, nodes on the same directed path are ancestors or descendants even if not directly connected by an edge. In Figure 2b, nodes A and B are ancestors of D₃ although they are not parents. Similarly, Y is a descendant of D₃ but not a child.

An essential property of BNs is the Markov property. If we index all the nodes in the DAG of BN as $Z = \{Z_1, Z_2, \dots, Z_p\}$, the joint probability distribution of P_z can be written as Equation (1):

$$P_z(Z) = \prod_{i=1}^p P_{Z_i}\{Z_i|PA(Z_i)\} \quad (1)$$

Here, we use $PA(Z_i)$ to designate the set of parent nodes of Z_i . This property greatly facilitates computation of probabilities regarding any subset of variables on the graph.

Given that both AOP networks and BNs share the structure of DAGs, it is natural to build BNs on AOP networks and use them to predict the adverse outcomes using the measurements regarding KEs in AOPs. AOPs suggest causal links between upstream KEs and downstream KEs or outcomes. In BNs, the joint probability distribution is written as the product of the conditional distributions of child nodes given their parent nodes (Equation (1)). If the causal relationships of AOPs are valid, it is reasonable to assume that the probability distributions of downstream KEs

依赖于上游事件。因此，用贝叶斯网络（BNs）来建模AOP网络在概念上是很有吸引力的。如果AOP网络能够用已知参数的BNs建模，不仅可以 根据前置关键事件（KEs）的信息对不良结果进行推理，还可以在观察到后代节点的情况下进行‘逆向’推理。有不同的方法来设置BN模型中的参数。公开发表的研究方法包括使用专家知识和从整理的数据集中学习参数。将贝叶斯框架应用于建模的一个重要优势是可以自然地量化模型的不确定性，这对于风险评估应用至关重要。

在这项研究中，我们关注的核心问题是预测毒性潜力作为顶端不良结果，也就是图2b中的Y。原则上，可以通过使用图2b中的整个网络计算与Y相关的概率来实现这一点。然而，我们将展示，通过利用BN的一些基本性质，这一任务可以大大简化。为此，我们引入了BN中的d-分离的概念，这是一种用于判断在给定第三组变量的情况下，两组变量是否独立的标准。读者可以查阅相关文献^{(p36),(p37)}了解d-分离的技术讨论。从概念上讲，如果在DAG中，一组节点W d-分离了集合U和集合V，那么在给定集合W的情况下，集合U和集合V在统计上是独立的。这意味着，如果我们拥有集合W的信息，集合U不会为集合V提供额外信息，反之亦然。因此，我们可以尝试找到一组方便测量的W，用于推断V，而无需担心集合U中的节点。如何i

只允许节点是离散的或服从正态分布。这有时会导致误解，认为贝叶斯网络只能应用于离散或正态随机变量。实际上，贝叶斯网络中的随机变量可以采用多种概率分布。然而，对于更一般的分布，精确计算概率通常变得不切实际，此时马尔可夫链蒙特卡罗（MCMC）模拟成为首选的实现方法。这可以通过通用的贝叶斯建模软件包来完成，例如JAGS。^{(p38),(p39),(p40)} 在我们的示例中，我们使用JAGS来实现一个稀疏贝叶斯模型，从而获得卓越的性能。

药物肝脏毒性预测：案例研究

在这里，我们以药物肝毒性为例，说明如何利用贝叶斯网络（BNs）的特性来促进不良结果预测模型的构建。这也提供了一个机会，可以经验性地评估AOP网络的模型呈现是否与BNs的数学特性一致。药物相关的肝毒性是药物开发和批准过程中的主要安全关注之一。因此，在描述与肝毒性相关的AOP方面已经付出了大量努力。

肝脏毒性AOP网络

我们构建了一个用于模拟药物肝毒性的AOP网络（图3），使用了从AOPwiki（<https://aopwiki.org/>）和文献中收集的与肝脏相关的AOPs。药物相关的肝毒性涉及许多生物过程，相应的AOP网络可以根据所选择的粒度以多种方式呈现。因此，我们构建了以已发表的AOPs为基础的AOP网络，强调具有可用体外检测方法的分子事件，以说明BN建模；其目的并非对药物相关肝毒性机制进行详尽的考察。图3中的粉色椭圆用于将不同的AOP分组以便于理解，并未在模型中使用。为简洁起见，图3仅包括我们已获得数据的分子事件。模型中还有许多其他关键事件（KEs）可被纳入，如胆红素尿和甘油三酯积累。我们省略了图3中粉色椭圆以外的大多数这些KEs，因为它们没有被

BN 模型

鉴于与基因表达相关的随机变量是连续的，我们假设它们服从正态分布，其均值是其父节点的线性函数。对于响应变量肝毒性，这是一个二元变量，我们假设采用逻辑回归模型（公式 (2)：

$$\log \frac{P(Y = 1|\mathbf{X} = \mathbf{x})}{P(Y = 0|\mathbf{X} = \mathbf{x})} = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}, \tag{2}$$

其中 Y 表示肝毒性潜力（0 表示无，1 表示有），向量 X 为 Y 的所有父节点的数值。在图 3 中，Y 的父节点为 27 个黄色节点。总的来说，与图 3 中 AOP 网络相关的贝叶斯网络包含 42 个节点，包括响应变量肝毒性潜力（Y）。如前所述，粉色椭圆表示胆汁淤积、脂肪变性、稳态失衡

depend on upstream events. Thus, modeling AOP networks with BNs is conceptually appealing. If AOP networks can be modeled with BNs with known parameters, one can perform not only inference on adverse outcomes based on the information of preceding KEs, but also ‘reverse’ inference on events given the observations on the nodes that are descendants. There are different ways to set the parameter in the BN model. Approaches in published work include using expert knowledge and learning the parameters from a curated data set. One important advantage of applying the Bayesian framework for modeling is that model uncertainty can be quantified naturally, which is vital for risk assessment applications.

In this study, we focus on the problem of predicting toxicity potential as the apical adverse outcome, that is, Y in Figure 2b. In principle, one can achieve this by computing the probability associated with Y using the entire network in Figure 2b. However, we will show that this task can be simplified substantially by leveraging some basic properties of BNs. For this, we introduce the idea of d-separation in BN, which is a criterion for determining whether two sets of variables are independent given a third set. Readers can consult relevant sources^{(p36),(p37)} for the technical discussion of d-separation. Conceptually, if a set ***W*** of nodes d-separate set ***U*** and set ***V*** in a DAG, set ***U*** and set ***V*** are statistically independent given set ***W***. This means that, if we have information on set ***W***, set ***U*** will provide no additional information for set ***V***, and vice versa. Thus, we can try to find a set ***W*** that is convenient to measure for the inference of ***V*** without worrying about nodes in ***U***. How to identify the sets for d-separation is discussed in the references given above. However, here we focus on sets that d-separate the end outcome (node Y) from some of its ancestors (e.g., node A). For this, we can simply find a set of nodes that block every directed path of arcs from node A to node Y. For example, ***C***= {*C*₁, *C*₂, *C*₃} is such a d-separation set. Similarly, ***X***= {*X*₁, *X*₂, *X*₃} is also a d-separation set. The practical implication for modeling is that, if we have measurements on nodes in ***C***, nodes A and B will not provide any further useful information for the inference of Y. Similarly, if nodes in ***X*** were measured, all the nodes above ***X*** will not provide useful information regarding Y. This property of d-separation can be used to construct more sparse models for practical applications, as we show in our example in the next section. The set ***X***, especially, is a Markov blanket (see references above) regarding Y, which is the minimal set of nodes that provides all the useful information for Y for all other nodes that are above it.

Although AOP networks and BNs share the underlying structure of DAG, whether AOP networks are in fact consistent with BNs and have the Markov property is an assumption that one needs to make for the model. In the next session, we show that, for our example, the result based on the liver toxicity AOP network is congruent with the mathematical property of BNs.

Finally, we want to comment on the computational aspects of BN models. When a BN is large, making inference involves computation over a large number of nodes, which could be computationally intensive. Efficient algorithms exist when all the nodes represent discrete random variables or continuous variables following a normal distribution.^(p37) Thus, many software packages

only allow nodes to be discrete or normally distributed. This sometimes leads to the misconception that BNs can only be applied to discrete or normal random variables. In fact, random variables in BNs can take a range of probability distributions. However, with more general distributions, exact calculation of probabilities often becomes impractical, and Markov chain Monte Carlo (MCMC) simulation becomes the preferred method for implementation. This can be carried out with general purpose Bayesian modeling packages, such as JAGS.^{(p38),(p39),(p40)} In our example, we use JAGS to implement a sparse Bayesian model that leads to superb performance.

Prediction for drug liver toxicity, a case study

Here, we use drug liver toxicity to illustrate how to utilize the properties of BNs to facilitate model building in the prediction of adverse outcomes. It also provides an opportunity to empirically evaluate whether model presentation of AOP networks is congruent with the mathematical properties of BNs. Drug-related liver toxicity is one of the major safety concerns in drug development and approval.^{(p41),(p42)} Accordingly, there has been significant effort to describe AOPs related to liver toxicity.^{(p43),(p44),(p45),(p46),(p47),(p48)}

AOP networks for liver toxicity

We built an AOP network for modeling drug liver toxicity (Figure 3), using liver-related AOPs collected in AOPwiki (<https://aopwiki.org/>) and from the literature.^{(p24),(p43),(p49),(p50)} Drug-related liver toxicity involves many biological processes and the corresponding AOP networks can be presented in multiple ways depending on the chosen granularity. Thus, we built AOP networks with published AOPs emphasizing molecular events that have available *in vitro* assays to illustrate BN modeling; it is not meant to be an exhaustive examination of mechanisms of drug-related liver toxicity. The pink ovals in Figure 3 are used to group different AOPs for clarity and are not used in the model. For conciseness, Figure 3 only included molecular events for which we had obtained data. There are numerous other KEs that could be included in the model, such as bilirubinuria and triglyceride accumulation. We omitted most of these KEs except those represented by pink ovals in Figure 3 because they were not used in the BN model er to keep the graph at a reasonable size.

BN models

Given that the random variables associated with gene expression are continuous, we assumed that they had a normal distribution, with the mean being a linear function of its parent nodes. For the response, hepatotoxicity, which is a binary variable, we assumed a logistic model (Equation (2)：

$$\log \frac{P(Y = 1|\mathbf{X} = \mathbf{x})}{P(Y = 0|\mathbf{X} = \mathbf{x})} = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}, \tag{2}$$

where Y is the hepatotoxicity potential (0 means no, 1 means yes) and the vector ***X*** is the value on all the parent nodes of Y. In Figure 3, the parent nodes of Y are the 27 nodes in yellow. In total, the BN associated with AOP network in Figure 3 contains 42 nodes, including the response, hepatotoxicity potential (Y). As mentioned earlier, the pink ovals (cholestasis, steatosis, dyshome-

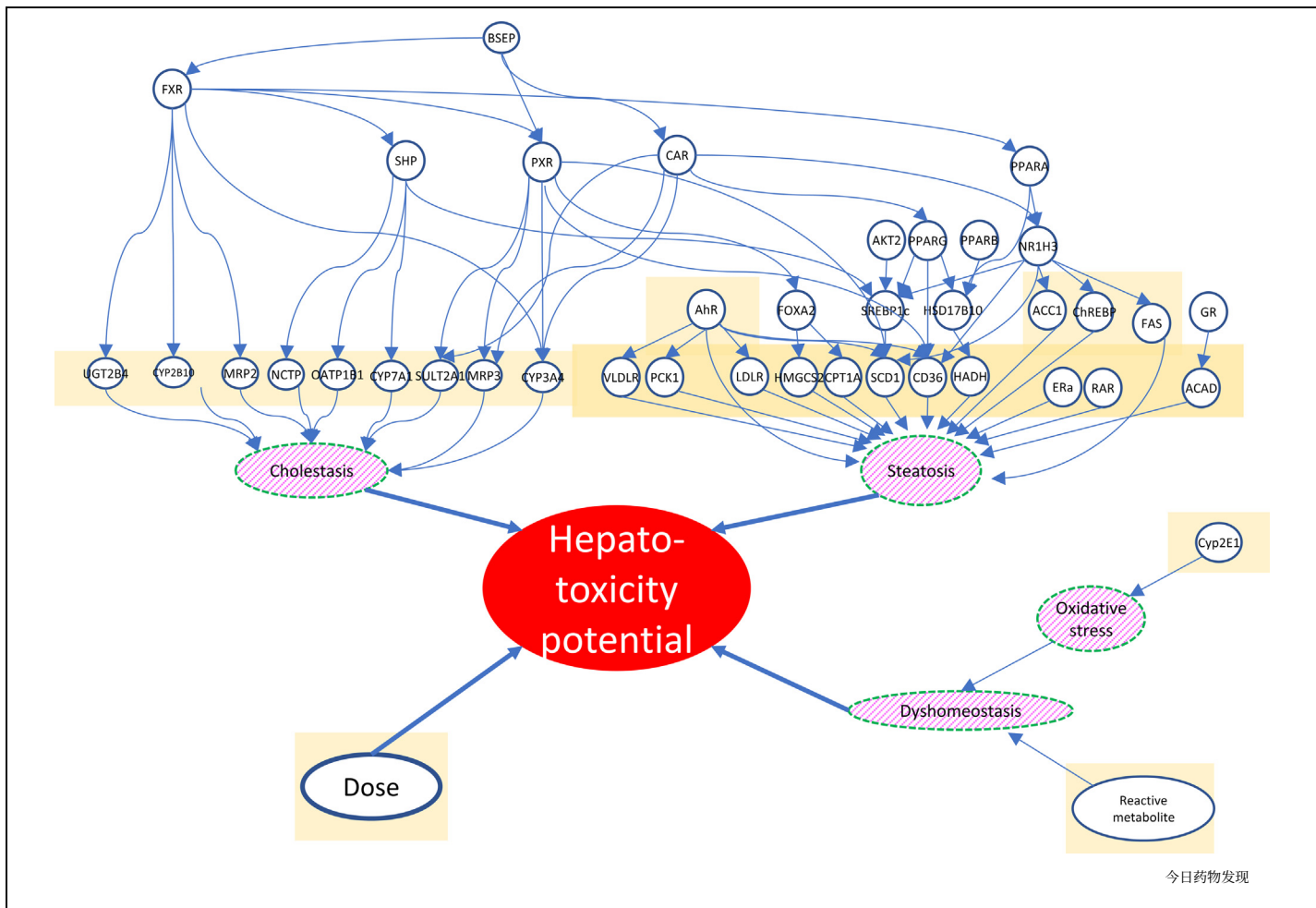


图3

用于药物肝毒性潜力的不良结果通路 (AOP) 网络。粉色椭圆用于叙述清晰, 并未包含在模型中。黄色区域内的节点构成肝毒性潜力的马尔可夫包络。我们仅在此图中包含有数据的分子事件; 本图并非对肝毒性相关AOP的全面展示。(关于本图例中色彩的解释, 请参阅本文的网络版本。)

稳态和氧化应激) 仅为了机制上的清晰而提出, 并不是BN模型的一部分。

根据前一节中概述的概念, Y 的父节点构成了肝毒性潜力的马尔可夫毯。当我们拥有这 27 个变量的值时, 其他节点不会为预测 Y 的状态提供额外信息。我们可以使用方程 (2) 中的模型推断肝毒性潜能存在的概率 $P(Y = 1)$ 。为了完整指定贝叶斯模型, 我们为 $b = b_k$ $p \ k \ 1$ 中的每个回归系数采用 horseshoe 先验^{(p55),(p56),(p57)}。我们假设 b_k 的先验分布为正态尺度混合 (方程 (3)):

$$\beta_k | \gamma_k, \tau_k \sim N(0, \gamma_k^2 \tau_k^2), \gamma_k \sim C^+(0, 1) \quad (3)$$

其中 c_k 是局部收缩参数, s 是整体全局收缩参数, C 是半柯西分布。Carvahlo 及其同事^(p58) 表明, 对于表示纯噪声且没有有用信息的变量, horseshoe 先验可以将其系数收缩到零。对于对响应有真实影响的变量, 对系数的收缩效果

马蹄铁先验的影响是温和的。我们的经验表明, 在这种情况下采用马蹄铁先验是有利的。如果不对系数进行适当的收缩, 模型可能变得不稳定且性能较差。我们使用 JAGS^(p38) 和 Rjags 包 (<https://cran.r-project.org/package=rjags>) 通过 MCMC 执行该贝叶斯模型的推断。如果将分布类型限制为离散或正态, 则无法应用马蹄铁先验。因此, 使用像 JAGS 这样的通用贝叶斯建模软件对于拟合更灵活的 BN 模型是必不可少的。

不同模型的结果和比较

我们基于图3所示的AOP网络 (或其子集) 拟合了几个BN模型, 以利用上述数据获得其参数。然后使用BN计算药物具有肝毒性潜力的概率; 根据分子实验的结果, 概率大于或等于 0.5 的药物被认为具有肝毒性潜力。我们将结果与DILI评分^(p51) 的分类进行比较, 以计算敏感性、特异性和准确性。作为公式(1)和马尔可夫的结果

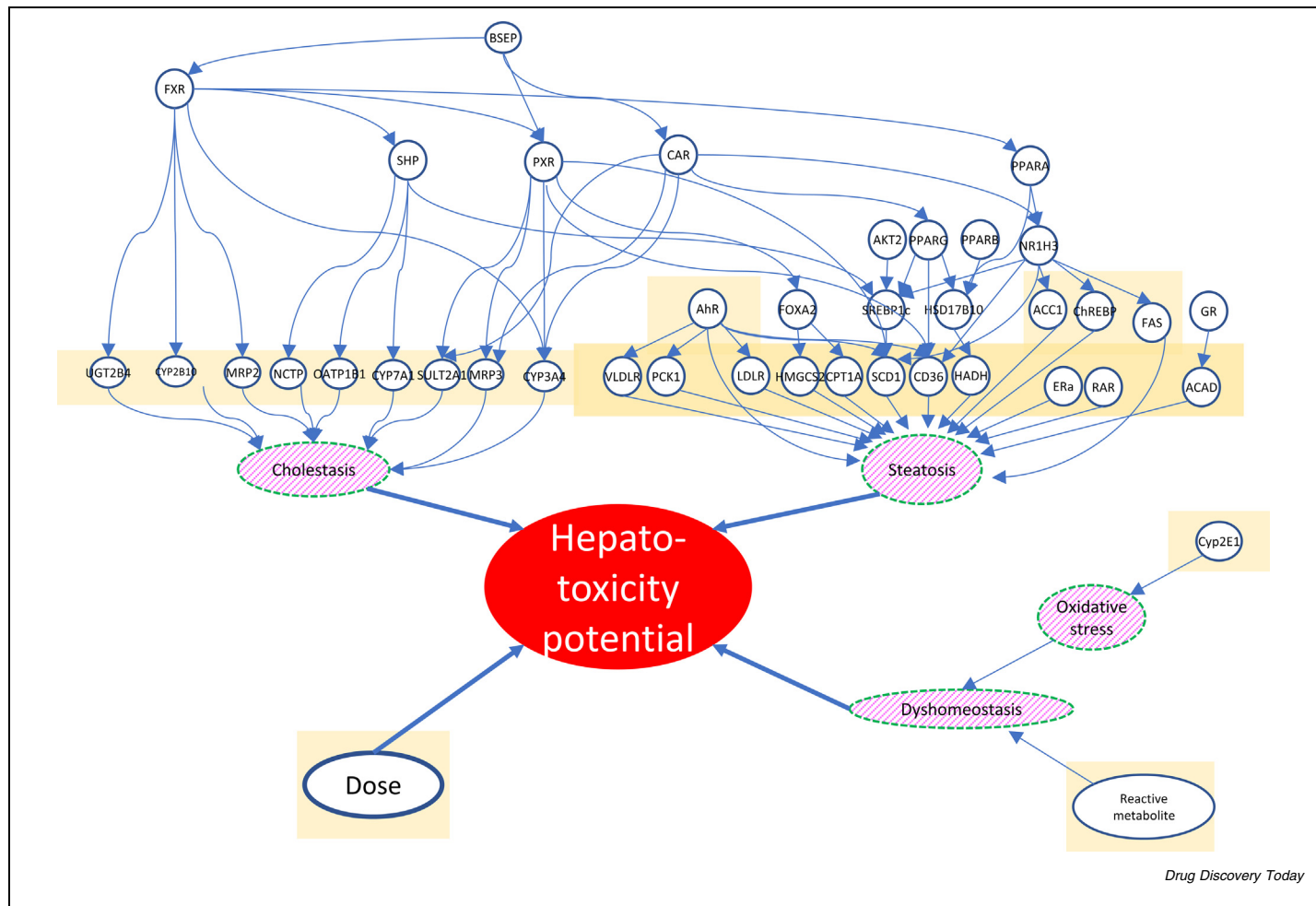


FIGURE 3

An adverse outcome pathway (AOP) network for drug hepatotoxicity potential. The pink ovals are presented for narrative clarity and are not included in the model. Nodes in the yellow area form the Markov blanket of hepatotoxicity potential. We only included molecular events with available data in this graph; it is not meant to be an exhaustive presentation of liver toxicity-related AOPs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ostasis, and oxidative stress) are presented solely for mechanistic clarity and are not part of the BN model.

Following the concepts outlined in the previous section, the parent nodes of Y form a Markov blanket for hepatotoxicity potential. When we have values for these 27 variables, other nodes do not provide additional information for predicting the state of Y . We can infer the probability of the presence of hepatotoxicity potential, $P(Y = 1)$, with the model in Equation (2). To fully specify the Bayesian model, we adopt horseshoe priors^{(p55),(p56),(p57)} for each regression coefficient in $\beta = \{\beta_k\}_{k=1}^p$. We assume that β_k has a prior distribution of scale mixture of normal (Equation (3)):

$$\beta_k | \gamma_k, \tau_k \sim N(0, \gamma_k^2 \tau_k^2), \gamma_k \sim C^+(0, 1) \quad (3)$$

where γ_k is a local shrinkage parameter, τ is an over-all global shrinkage parameter, and C^+ is the half-Cauchy distribution. Carvahlo and coworkers^(p58) showed that, for variables representing pure noise without useful information, the horseshoe priors can induce shrinkage of their coefficients to zero. For variables with a real effect on the response, the shrinkage effect on coefficients

by the horseshoe prior is mild. Our experience showed that adoption of the horseshoe prior in this setting was advantageous. Without proper shrinkage of the coefficients, the model can become unstable with inferior performance. We used JAGS^(p38) and the Rjags package (<https://cran.r-project.org/package=rjags>) to perform inference for this Bayesian model with MCMC. Application of the horseshoe prior was unfeasible if we limited the types of distribution to be either discrete or normal. Thus, using a generic Bayesian modeling software such as JAGS is essential to fit more flexible models for BN.

Results and comparisons for different models

We fitted several BN models based on the AOP network in Figure 3 (or its subsets) to obtain their parameters with the data described above. The probability for the drug to have hepatotoxicity potential was then computed with the BN; drugs with probability >0.5 based on molecular assays were considered to have hepatotoxicity potential. We compared the result with the classification of DILI score^(p51) to calculate sensitivity, specificity, and accuracy. As a consequence of Equation (1) and the Markov

表格 1							
多种BN模型对药物肝毒性潜力的结果							
模型	真正例	误报	真正的阴性	假阴性	敏感度	特异性	准确性
马尔可夫毯/完整	84	9	45	8	0.903	0.849	0.884
Flat	84	9	45	8	0.903	0.849	0.884
上套	83	12	42	9	0.874	0.823	0.856
上部分集	83	22	32	9	0.790	0.780	0.788

正如之前讨论的BN特性一样，肝毒性潜力Y的概率分布完全由马尔可夫毯子中的变量值决定。因此，如果我们只对肝毒性潜力的预测感兴趣，那么基于图3中整个网络（41个节点加响应）的BN与仅基于马尔可夫毯子中的节点（黄色阴影的27个节点）的BN给出了相同的结果。因此，我们在表1中以单行（马尔可夫毯子/完整）展示结果。它们给出的敏感性为0.903，特异性为0.849，准确率为0.884（表1）。

如果可以合理地将AOP网络建模为BN，这意味着我们只需要使用马尔可夫毯中的子节点（27个节点）进行建模，而无需使用所有节点（41个节点），这将大大减轻计算负担并且需要的实验也更少。然而，这是一个尚未经过实证评估的假设。为了评估AOP网络是否与BN的数学特性一致，我们还拟合了一个“平坦”模型（图4a）。平坦模型是一个逻辑回归模型，每个体外实验都可以影响肝毒性风险，并且每个回归系数都采用马蹄形先验。如果图3中的AOP网络不满足马尔可夫性质，并且马尔可夫毯之外的节点可以提供肝毒性潜力的有用信息，那么平坦模型的表现应优于基于马尔可夫毯的模型或完整的BN模型。表1（前两行）中的结果显示，实际上它们的表现是相同的，这证实了AOP网络是符合…

利用马尔可夫性质，仅使用马尔可夫毯中的节点构建的BN模型可以充分利用AOP网络中包含的所有信息。然而，有时可能更希望找到一个节点数比马尔可夫毯更少的子集。这对于降低体外实验的成本和物流负担很有用，但代价是由于信息丢失而导致模型性能下降。为此，我们拟合了图4b中绿松石区域节点的“上集”模型。该集合包含九个节点，少于马尔可夫毯的27个节点。该节点集合的一个特性是它们形成了响应（肝毒性潜力）与MIEs之间的d分离集；也就是说，从MIEs到响应的所有有向路径都必须经过该集合中的某一个节点。表1（第3行）显示，与基于马尔可夫毯节点的模型相比，这个上集模型的性能有所下降（敏感性0.874，特异性

我们还探索了一个我们称之为“上部分集合”模型的模型。上部分集合包括上集中的节点，但不包括FXR和反应性代谢物节点；因此，该集合有七个节点。在这里，FXR是几个下游基因的祖先

流和反应性代谢物是药物的属性。移除这两个节点打破了起始事件与最终反应之间的d-分离，因为存在通过这两个被排除节点从起始事件到反应的路径。表1（第4行）显示，该模型的性能更明显下降（敏感性0.790，特异性0.780，准确性0.788），尤其是在特异性方面。

讨论
AOP 是一个有价值的框架，用于与毒性风险评估相关的知识组织。鉴于 AOP 网络与 BN 共享基础的 DAG 结构，将这两个概念结合起来进行推理任务是很自然的。使用基于 AOP 的 BN 进行风险评估已显示出一定的成功。(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23) 然而，作为具有特定假设和数学特性的统计模型，BN 在 AOP 网络背景下尚未被仔细评估。因此，我们使用肝毒性 AOP 网络来经验性地评估马尔可夫性质是否适用于 AOP 网络。我们的结果与马尔可夫毯中的节点包含预测反应（肝毒性潜力）所需的所有信息相一致，这支持了在基于 AOP 的建模中使用 BN。这也表明，可以优先考虑马尔可夫毯中的节点进行预测，从而显著减少所需的体外实验数量。
Burgoon

AOP 网络的结构还提供了额外的信息，可用于促进模型构建和测定开发。我们探索了使用除马尔可夫毯以外的节点子集。上层节点集的节点较少（九个对比 27 个），但它仍然在起始事件与响应之间形成了 d 分离。使用上层节点集的模型相比使用马尔可夫毯节点的模型性能略有下降。当 d 分离特性被破坏时，例如使用上层部分节点集时，性能下降会更加严重。这指出了一种方法，即选择较少的测定以降低成本，同时接受模型性能的轻微下降。正如一位审稿人指出的，AOP 中“上游”的节点通常代表较低层次的生物组织（细胞或组织），在较低层次生物组织进行的测定在金钱和时间上的成本可能也较低。它也可能

TABLE 1							
Results for several BN models for drug hepatotoxicity potential							
Model	True positive	False positive	True negative	False negative	Sensitivity	Specificity	Accuracy
Markov blanket/Full	84	9	45	8	0.903	0.849	0.884
Flat	84	9	45	8	0.903	0.849	0.884
Upper set	83	12	42	9	0.874	0.823	0.856
Upper partial set	83	22	32	9	0.790	0.780	0.788

property for BN as discussed previously, the probability distribution of the hepatotoxicity potential, Y, was entirely determined by the values of variables in the Markov blanket. Thus, the BN based on the entire network in [Figure 3](#) (41 nodes plus the response) gave the same result as the BN based only on the nodes in the Markov blanket (the 27 nodes in the yellow shade) if we were only interested in making predictions for the hepatotoxicity potential. Thus, we show the results in a single row (Markov blanket/Full) in [Table 1](#). They gave a sensitivity of 0.903, specificity of 0.849, and accuracy of 0.884 ([Table 1](#)).

If AOP networks can be reasonably modeled as BNs, it means we only need to use the subset of nodes (27 nodes) in the Markov blanket for modeling rather than all the nodes (41), which makes the computation burden much smaller and requires fewer assays. However, this is an assumption that has not been empirically evaluated. To assess whether AOP networks are congruent with the mathematical properties of BN, we also fitted a flat’ ([Figure 4a](#)). The flat model is a logistic model where every *in vitro* assay can affect the risk of liver toxicity with the horse-shoe priors on each regression coefficient. If the Markov property does not hold for the AOP network in [Figure 3](#) and nodes outside of the Markov blanket can provide useful information for the hepatotoxicity potential, the flat model should perform better than the Markov blanket-based model or the full BN model. Results in [Table 1](#) (first two rows) shows that the performances were, in fact, the same, confirming that AOP networks are congruent with the Markov property assumption.

With the Markov property, the BN model using only nodes in the Markov blanket fully utilizes all the information contained in the AOP network. However, it might sometimes be preferable to find a subset that has an even smaller number of nodes than the Markov blanket. This is useful for reducing the cost and logistic burden for performing *in vitro* assays at the expense of reduced model performance because of the lost information. For this, we fitted the ‘upper set’ model with nodes in the turquoise region in [Figure 4b](#). This set included nine nodes, fewer than the 27 for the Markov blanket. One property of this set of nodes is that they formed a d-separation set between the response (hepatotoxicity potential) and the MIEs; that is, all the directed paths from the MIEs to the response had to pass through one of the nodes in this set. [Table 1](#) (Row 3) shows that this upper set model had reduced performance compared with that based on nodes in the Markov blanket (sensitivity 0.874, specificity 0.823, and accuracy 0.856), although the differences are not dramatic.

We also explored a model that we called the ‘upper partial set’ model. The upper partial set included the nodes in the upper set except the nodes FXR and reactive metabolite; thus, this set had seven nodes. Here, FXR is the ancestor of several genes down-

stream and reactive metabolite is a drug property. Removing these two nodes broke down the d-separation between the initiating events and the final response because there were paths from the initiating events to the response through the two excluded nodes. [Table 1](#) (Row 4) shows that the performance of this model deteriorated more significantly (sensitivity 0.790, specificity 0.780, and accuracy 0.788), especially for specificity.

Discussion

AOP is a valuable framework for knowledge organization related to toxicity risk assessment. Given that AOP networks share the underlying DAG structure with BNs, it is natural to perform inference tasks by combining these two concepts. Using AOP-based BNs for risk assessment has shown some success.(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23) However, as statistical models, with specific assumptions and mathematical properties, BNs have not been carefully evaluated in the context of AOP networks. Thus, we used liver toxicity AOP networks to empirically assess whether the Markov property holds for AOP networks. Our results are congruent with the nodes in the Markov blanket containing all the information to predict the response (hepatotoxicity potential), which supports the use of BNs for AOP based modeling. This also suggests that one can prioritize nodes in the Markov blanket for prediction, thus significantly reducing the number of *in vitro* assays that are required. Burgoon *et al.* ^(p16) presented an approach based on the backdoor algorithm (Pearl1993) ^(p59) with the same effect. The concepts of Markov blanket and d-separation that we present here provide a more convenient framework to identify a subset of nodes (assays).

The structure of AOP networks also provides additional information that can be utilized to facilitate model building and assay development. We explored using a subset of nodes other than the Markov blanket. The upper set of nodes had fewer nodes (nine versus 27), but it still formed d-separation between the initiating events and the response. The model using the upper set had a moderately decreased performance compared with that using nodes in the Markov blanket. When the d-separation property was violated, such as with the upper partial set, the deterioration of performance became more severe. This points to an approach to select a smaller set of assays to reduce cost while accepting a mild reduction in model performance. As one reviewer pointed out, nodes ‘upstream’ in an AOP typically represent lower levels of biological organization (cells or tissues) and assays performed at lower levels of biological organization can be less costly in both money and time. It might also reduce the use of animal for drug testing. These are additional arguments to support the idea that a smaller set of assays closer to the MIE might be optimal. When feasible, a set of nodes that cause d-separation

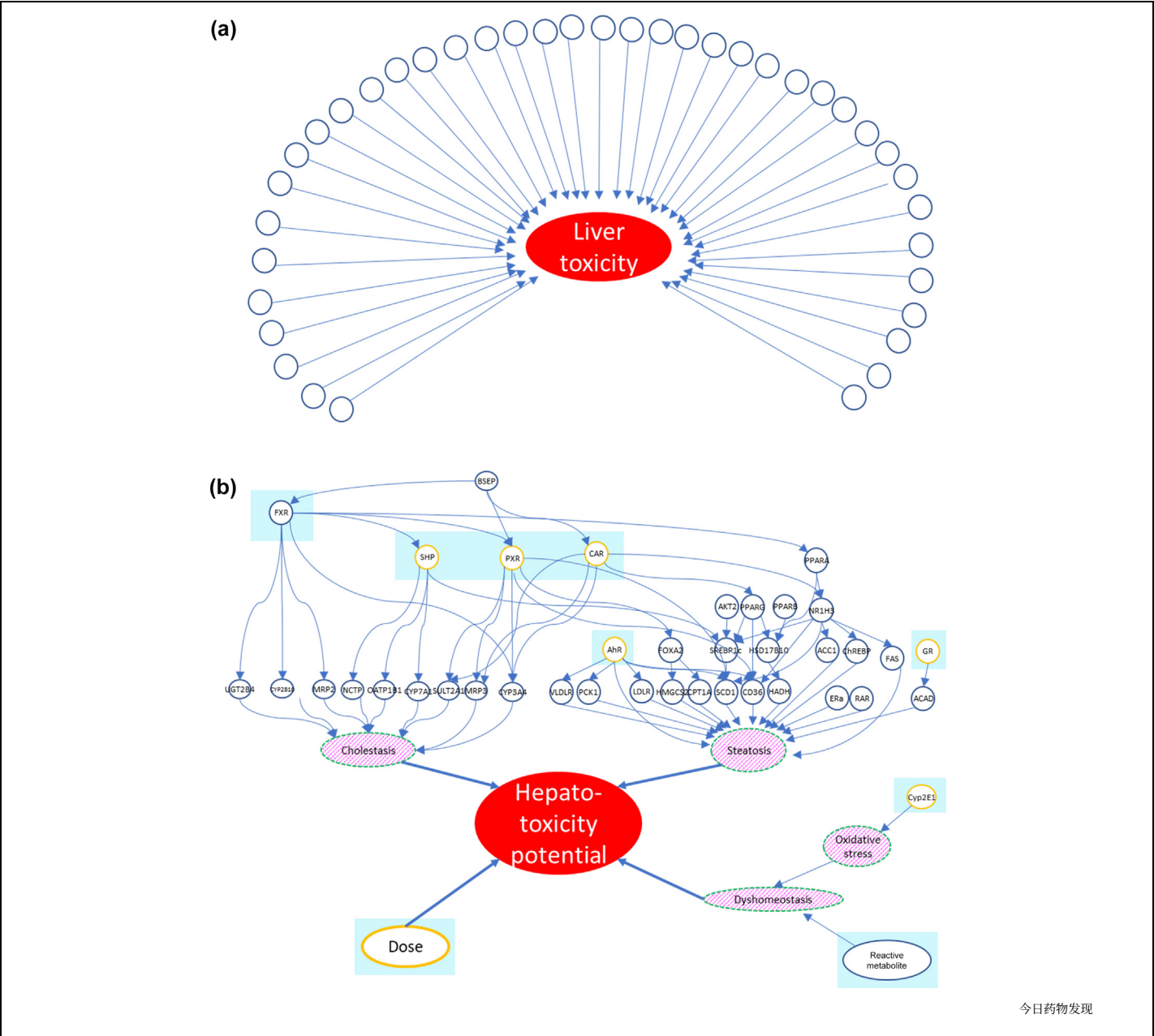


图 4
平面模型和部分上集合模型的示意图。(a) 使用平面模型的贝叶斯网络 (BN)。所有41个变量都可以为肝毒性风险的预测提供信息。(b) 上集合的示意图。上集合中的节点位于绿松石色区域。

在初始事件与响应之间的关系往往能够提升模型的性能。

在我们的肝脏毒性贝叶斯模型中，我们对系数采用了马蹄铁先验，这对模型的良好性能至关重要。这表明，在贝叶斯网络中使用广泛的分布类型，而不仅限于离散和正态分布，是有优势的。为了允许使用广泛的分布类型，需要通用的贝叶斯建模软件，例如我们示例中使用的 JAGS。在贝叶斯建模框架中，还可以考虑更复杂的模型。例如，在指定 KE 节点及其父节点时，我们假设基因表达水平的均值呈线性关系。该模型在案例研究中给出了令人满意的结果。然而，非线性关系或

非参数形式可以通过现代贝叶斯统计软件实现。像任何建模工作一样，需要在更灵活模型的能力与模型复杂性的代价之间取得平衡，这一点可以在未来的研究中进一步探讨。我们的例子还表明，在药物开发过程中结合 AOP 和 BN 具有潜力。鉴于 AOP 正在持续发展，而 BN 是一种高度强大的统计方法，因此预计将在结合这两种方法方面取得更多进展。

最后，与任何模型一样，贝叶斯网络 (BNs) 涉及一定程度的抽象。在我们讨论 AOP 网络与 BNs 之间的关系时，某些属性只有在我们的知识是完美的情况下才会严格成立，也就是说，AOP 网络能够封装

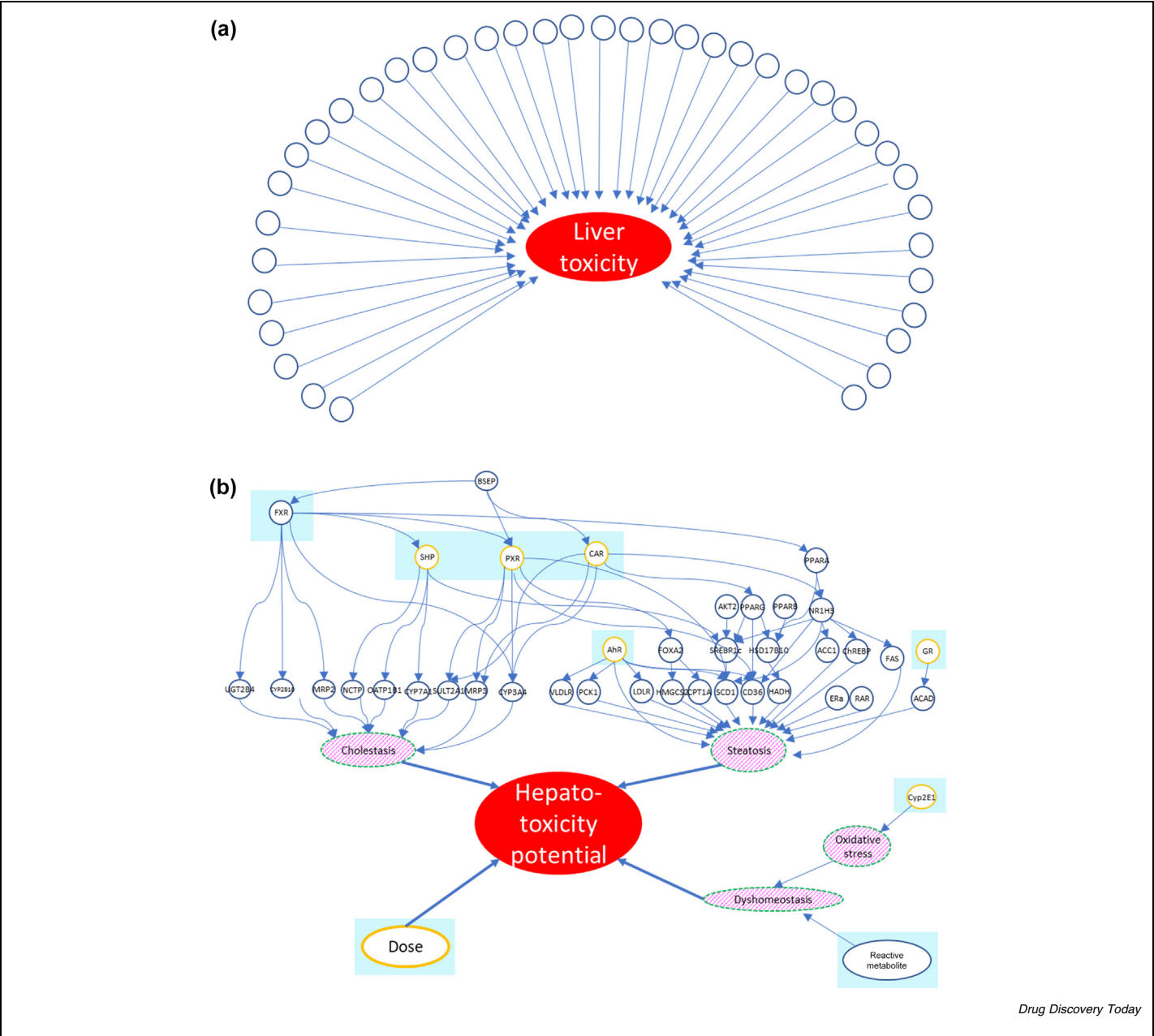


FIGURE 4
Illustration of the flat model and the partial upper set model. **(a)** The Bayesian network (BN) with the flat model. All 41 variables can provide information for the prediction of liver toxicity risk. **(b)** Illustration for the upper set. Nodes in the upper set are in the turquoise area.

between the initiating events and the response tends to enhance model performances.

In our Bayesian model for liver toxicity, we adopted horseshoe priors for the coefficients, which was crucial for good model performance. This suggests the advantage of using a broad range of distribution types in BNs rather than being limited to discrete and normal distributions. To allow a broad range of distributions, a generic Bayesian modeling software is required, such as JAGS used in our example. In the Bayesian modeling framework, even more sophisticated models can be considered. For example, we assumed a linear relationship for the mean of gene expression levels in specifying the KE node given its parent nodes. This model gave satisfactory results in the case study. However, nonlinear relationships or

nonparametric forms could be implemented with modern Bayesian statistical software. As in any modeling exercise, one needs to balance the power of more flexible models with the cost of model complexity, an aspect that could be explored further in future studies. Our example also demonstrated that there is potential in combining AOP and BN in the drug development process. Given that AOPs are being continuously developed and BN is a highly powerful statistical approach, more advances that combine these two approaches are expected.

Finally, as with any model, BNs involve certain levels of abstraction. In our discussions on the connection between AOP networks and BNs, some of the properties will strictly hold only if our knowledge is perfect, that is, the AOP networks encapsulate

导致不良结果的所有可能路径。这一假设在实践中可能并不总是成立，因为生物学的复杂性非常庞大。当高分辨率数据可用时，就有可能发现从上游节点到不良结果途径（AOP）的新路径和新关系。因此，模型应随着新的生物学知识不断发展，类似于AOP的持续演变。

作者贡献

D.W.: 撰写 – 审稿与编辑；撰写 – 原稿；调查；数据整理；概念化。
A.S.: 撰写 – 审稿与编辑；撰写 – 原稿；验证；概念化。
W.T.: 撰写 – 审稿与编辑；概念化。

数据可用性

肝毒性分类的数据可通过肝脏毒性知识库获得（<https://www.fda.gov/science-research/bioinformatics-tools/liver-toxicity-knowledge-base-ltkb>）。L1000 项目的基因表达数据可从 <https://lincsproject.org/LINCS/tools/workflows/find-the-best-place-to-obtain-the-lincs-l1000-data> 获取。

Data

作为图 3 中 DAG 的 BN 模型的响应变量，药物肝毒性潜力来自肝脏毒性知识库（LTKB）。LTKB 是美国食品药品监督管理局（FDA）国家毒理研究中心的一个项目，专注于药物性肝损伤（DILI），提供了多个关于药物肝毒性的基准数据集。Chen 及其同事实现了 DILI 评分，一种针对 354 种药物（124 种最关注，162 种较少关注，68 种无需关注）肝毒性潜力的分类算法。

在 FDA 标签、DILI 案例注册、出版物和其他来源中。在我们的模型中，我们使用了最关注的药物（具有肝毒性潜力的药物或案例药物，N = 92）和无需关注的药物（没有明显肝毒性潜力的药物或对照药物，N = 54），这些药物在 LTKB 中具有可用的分子检测信息。对于与 AOP 网络中的分子事件相关的随机变量，我们使用每种药物处理下基因的归一化表达变化。L1000 检测的基因表达水平数据^(p9)来自基因表达综合数据库 (Gene Expression Omnibus)，登录号为 GSE92742 和 GSE70138。L1000 是一种高通量基于微珠的基因表达分析方法，可用于高效生成各种处理条件下细胞的全转录组谱。L1000 检测的数据提供了多种不同格式。为了进行分析，我们使用 CMapR 软件包^(p52) 提取了重复合并的 z 分数（Level 5 数据）并进行了进一步处理。

反应性代谢物形成的状态和剂量是从LTKB获得的。^(p51) 在这里，剂量是模型中唯一不是AOP网络中的分子事件或不良结果的随机变量。以往的研究表明，剂量是在预测药物肝毒性时的一个关键变量。^(p51)，^(p54)

数据可用性

数据可根据要求提供。

致谢

本文中表达的观点属于作者个人观点，并不一定反映 FDA 的立场。

利益冲突

作者声明没有利益冲突。

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all the possible pathways leading to the adverse outcome. This assumption might not always be true in practice,given the enormous complexity of biology. When high-resolution data become available, it will be possible to discover new pathways and relationships leading from upstream nodes to the AOP. Thus, models should evolve with new knowledge in biology, similar to the continuing evolvement of AOPs.

Author contributions

D.W.: writing – review & editing; writing – original draft; investigation; data curation; conceptualization. **A.S.:** writing – review & editing; writing – original draft; validation; conceptualization. **W.T.:** writing – review & editing; conceptualization.

Data availability

Data for hepatotoxicity classification are available through Liver Toxicity Knowledge Base (<https://www.fda.gov/science-research/bioinformatics-tools/liver-toxicity-knowledge-base-ltkb>). Gene expression data from the L1000 project can be obtained from <https://lincsproject.org/LINCS/tools/workflows/find-the-best-place-to-obtain-the-lincs-l1000-data>.

Data

As the response variable of the BN model with the DAG in [Figure 3](#), drug hepatotoxicity potential was obtained from the Liver Toxicity Knowledge Base (LTKB).^(p51) LTKB is a project of the US Food and Drug Administration (FDA)’s National Center for Toxicological Research that focuses on drug-induced liver injury (DILI), which provides several benchmark data sets regarding the liver toxicity of drugs. Chen and coworkers^(p51) implemented the DILI score, a classification algorithm of hepatotoxic potential for 354 drugs (124 most concern, 162 less concern, and 68 no concern) based

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on FDA labels, DILI case registries, publications, and other sources. In our model, we used most concern (drugs with hepatotoxic potential or case drugs, N = 92) and no concern (drugs with no evident hepatotoxic potential or control drugs, N = 54) drugs in LTKB for which molecular assay information was available. For random variables associated with molecular events in the AOP network, we use normalized expression change for the genes under treatment with each drug. Data for gene expression levels from L1000 assays^(p9) were obtained from the Gene Expression Omnibus with the accession numbers GSE92742 and GSE70138. L1000 is a high-throughput bead-based gene expression profiling assay that can be used to efficiently generate whole-transcriptome profiles for cells under various treatments. Data for L1000 assays were provided in several different formats. For our analysis, the replicate-collapsed z-scores (Level 5 data) were extracted with the CMapR package^(p52) and further processed as described previously.^(p53) The status of reactive metabolite formation and dose was obtained from LTKB.^(p51) Here, dose was the only random variable in the model that was not a molecular event or adverse outcome in the AOP network. Previous work has shown that dose is a critical variable in predicting drug liver toxicity.^{(p51),(p54)}

Data availability

Data will be made available on request.

Acknowledgment

The opinions expressed in this paper are those of the authors, and do not necessarily reflect the position of the FDA.

Conflicts of interest

The authors declare no conflicts of interest.

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